SOURCE CODE:

```
import tkinter as tk
from weather_api import
open_weather_window
from rainfall_analysis import
show_rainfall, analyze_india
from temperature import
predict_temp, compare_temp

def main():
    root = tk.Tk()
    root.title("Weather Intelligence
System")
    root.geometry("500x500")
    root.configure(bg="#1e272e")

    title = tk.Label(root,
text="Weather Intelligence System",
```

```
                font=("Arial", 18, "bold"),
                bg="#1e272e",
fg="white")
    title.pack(pady=20)

    tk.Button(root, text="Current
Weather",

command=open_weather_window,
width=30).pack(pady=10)

    tk.Button(root, text="Rain
Tomorrow Analysis",
            command=show_rainfall,
width=30).pack(pady=10)

    tk.Button(root, text="India
Rainfall Analysis",
```

```
            command=analyze_india,
width=30).pack(pady=10)

    tk.Button(root, text="Temperature
Prediction",
            command=predict_temp,
width=30).pack(pady=10)

    tk.Button(root, text="Compare
Temp (Actual vs Predicted)",
            command=compare_temp,
width=30).pack(pady=10)

    root.mainloop()

if __name__ == "__main__":
    main()import tkinter as tk
import requests
```

```
API_KEY = "YOUR_API_KEY"   # replace this

def get_weather(city):
    try:
        url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}"
        data = requests.get(url).json()

        if data["cod"] != 200:
            return "City not found"

        temp = int(data["main"]["temp"] - 273.15)
        condition = data["weather"][0]["main"]
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```
        return f"{condition}\n{temp}°C"

    except:
        return "Error fetching data"

def open_weather_window():
    win = tk.Toplevel()
    win.title("Weather")

    tk.Label(win, text="Enter
City").pack()

    entry = tk.Entry(win)
    entry.pack()

    result = tk.Label(win, text="")
    result.pack()
```

```
    def fetch():
        city = entry.get()
        output = get_weather(city)
        result.config(text=output)

    tk.Button(win, text="Get
Weather",
command=fetch).pack()import
pandas as pd
import matplotlib.pyplot as plt

def show_rainfall():
    df = pd.read_csv("data/
weatherAUS.csv")

    df['RainTomorrow'].replace({'No':
0, 'Yes': 1}, inplace=True)
```

```
df.RainTomorrow.value_counts(normalize=True).plot(kind='bar')
    plt.title("Rain Tomorrow Distribution")
    plt.show()

def analyze_india():
    df = pd.read_csv("data/india_rainfall.csv")

    df.mean().plot(kind='bar')
    plt.title("Average Monthly Rainfall in India")
    plt.show()import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection
```

```
import train_test_split
from sklearn.linear_model import
LinearRegression

df = pd.read_csv("data/
temperature.csv")
df.dropna(inplace=True)

X = df.iloc[:, :−1]
Y = df.iloc[:, −1]

def predict_temp():
    X_train, X_test, y_train, y_test =
train_test_split(X, Y, test_size=0.2)

    model = LinearRegression()
    model.fit(X_train, y_train)
```

```
    score = model.score(X_test,
y_test)

    print(f"Accuracy: {score * 100:.2f}
%")

def compare_temp():
    X_train, X_test, y_train, y_test =
train_test_split(X, Y, test_size=0.2)

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    plt.plot(y_test.values,
label="Actual")
    plt.plot(y_pred, label="Predicted")
```

```
    plt.legend()
    plt.title("Temperature Comparison")
    plt.show()
```

Conclusion:

Weather prediction plays a vital role in modern society, especially in sectors like agriculture, disaster management, transportation, and daily life planning. With the advancement of technologies such as IoT, machine learning, and cloud computing, weather forecasting has become more accurate, timely, and reliable. These systems collect real-time data from multiple sources

and analyze it to provide meaningful insights and predictions.

Accurate weather prediction helps farmers make better decisions regarding irrigation, crop protection, and harvesting, thereby improving productivity and reducing losses. It also helps governments and organizations prepare for extreme weather conditions like floods, cyclones, and droughts, minimizing damage and ensuring public safety.

Weather prediction is not just a scientific process but tool that supports sustainable elopment and risk management. Continuous

improvements.Accurate weather prediction helps farmers make better decisions regarding irrigation, crop protection, and harvesting, thereby improving productivity and reducing losses. It also helps governments and organizations prepare for extreme weather conditions like floods, cyclones, and droughts, minimizing damage and ensuring public safety.